



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering

Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: B.E., II Semester & ECE 'A'

Course Code & Title: EC3251 & Circuit Analysis

Name of the Faculty member (s): Dr.A.Lakshmi

Innovative Practice Description

- **Unit / Topic:** Unit 5 – Cut sets
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO 28
- **Activity Chosen:** Flipped Class
- **Justification:**

Complex circuits can be analyzed by converting them into network graphs. Graphs, branch link, tie set matrix are taught in the class. A cut-set is a minimum set of branches of a connected graph such that when removed these branches from the graph, then the graph gets separated into 2 distinct parts called sub-graphs and the cut set matrix is the matrix which is obtained by row-wise taking one cut-set at a time. Cut set is used to determine node voltage, hence those topics were chosen for the flipped class. Tie set matrix is similar to cut set matrix used to determine loop currents and branch currents which is already taught in the class. The pupils have the opportunity to explore the subject to learn how the operations are changed. This flipped class exercise was chosen as a result to get them thinking about cut set matrix representation.

- **Time Allotted for the Activity:** 45 minutes
- **Details of the Implementation:**

Students were given an outline of the flipped class activity before it began. For the Flipped Class activity, the students are provided with a self-created video which is shared through the Canvas (LMS). The video link contains the detailed description of text book problems and checking theoretical answers. The students are already divided into teams so when they come to class, they divide their tasks accordingly among themselves. The pupils were given question and internet resources about their subject. Ten minutes are allocated for presentation preparation. Each group will deliver the topic they have chosen, and the audience will listen. Following the presentation, there will be an opportunity for attendees to raise any questions or concerns they may have. This made it possible for every group to share in class the concepts they had learned about their individual subjects.

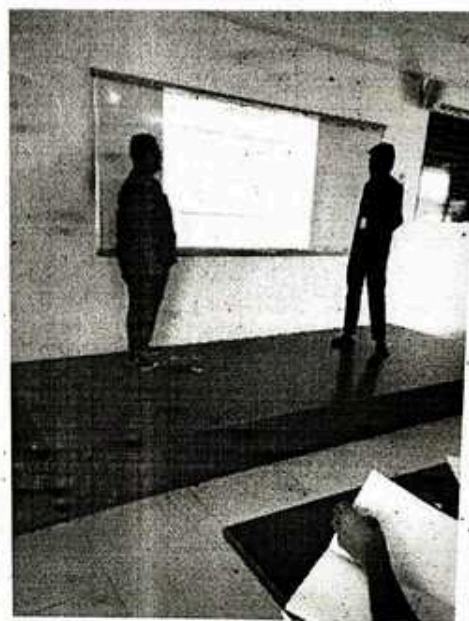
CO – PO / PSO mapping:

CO	PO5	PO9	PSO3
CO5	1	2	2

• **PO / PSO mapped:**

Innovative practice	PO9	PSO3
	2	2
Justification for correlation	The students will be able to explain the analyzing the node voltage using cut set matrix with effective presentations as individual or leader in diverse team hence it is mapped to level 2.	The students will be able to analyze the cut-set matrix which is used to solve any electric circuit or network problem by utilizing their equivalent graphs; hence it is mapped to level 2.

• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

The students' development of their presentation skills was aided by their exposure to a range of perspectives during group discussions and their capacity to apply theory to real-world situations. They were impressed by the creative thinking that went into the project and what could be accomplished with it. Educational materials could provide students with ideas for the presentation, assisting them in putting it together.

❖ **Benefit of the practice:**

Students successfully completed this task. They theoretically planned the presentation and set out the task for their chosen topic in a simple-to-understand

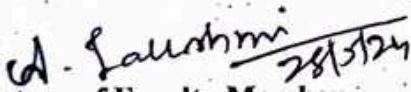
manner. Everyone in the group understood the topics well. Students who worked in groups were better able to comprehend subjects because they were allowed to share thoughts. Each group member was skilled at explaining the concepts of various cut-set matrix problems and how they may be used for calculating node voltage, current equation etc.

❖ **Challenges faced in implementation:**

- The presentation material was developed by some members without the relevant information.
- Some pupils had trouble delivering because there wasn't enough communication.
- A few students took longer than was allowed.
- Some students found it challenging to respond to the inquiries raised by other team members.

References:

1. Hayt Jack Kemmerly, Steven Durbin, "Engineering Circuit Analysis", Mc Graw Hill education, 9th Edition, 2018
2. Charles K. Alexander & Mathew N.O.Sadiku, "Fundamentals of Electric Circuits", Mc Graw- Hill, 2nd Edition, 2003
3. Joseph Edminister and Mahmood Nahvi, —Electric Circuits, Schaum's Outline Series, Tata McGraw Hill Publishing Company, New Delhi, Fifth Edition Reprint 2016.
4. <http://nitttrc.edu.in/nptel/courses/video/108105159/lec63.pdf>


Signature of Faculty Member


HOD



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: II Semester B.E. ECE A

Course Code & Title: EC3251 Circuit Analysis

Name of the Faculty member: Dr.A.Lakshmi

Innovative Practice Description

- **Unit / Topic:** Unit IV – The Unit – Step Function Driven RLC Circuits
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO 22
- **Activity Chosen:** Collaborative Learning- Team Assisted Individualization (TAI)
- **Justification:**

The Unit Step Function Driven RLC Circuits topic explained about the functionality of unit step functions in both series and parallel connected RLC circuits. Depends upon the switch position of RLC circuits the current/voltage can be Increasing/decreasing. The response of current/voltage can be drawn by the use of mathematical expression. So derivation of each case is very difficult to study individually, the TAI activity is very much helpful for understand the concept of RLC circuits. Team Assisted Individualization (TAI) refers to a teaching and learning approach that combines elements of both team collaboration and individualized instruction. The goal is to improve the learning experience to meet the unique needs and abilities of each learner while also fostering collaboration and teamwork.

Time Allotted for the Activity: 45 minutes

• Details of the Implementation:

➤ Group Formation:

Group the students based on their learning needs or interests for specific activities. (Combination of Bright, Average and below average students)

- **Introduction of Learning Material**
Faculty introduces a concept (RLC circuits – series and Parallel connection) or sharing the learning material to the whole class.
- **Individual Study and Preparation**
Each student in the group is encouraged to understand the concept of Unit step driven function in RLC circuits.
- **Use of Technology**
Faculty share the knowledge about integrate technology (online simulation) tool that support individualized learning (Falstad Simulator).
- **Monitoring and Adjustments**
Regularly monitor the effectiveness of the TAI approach through assessments and feedback.

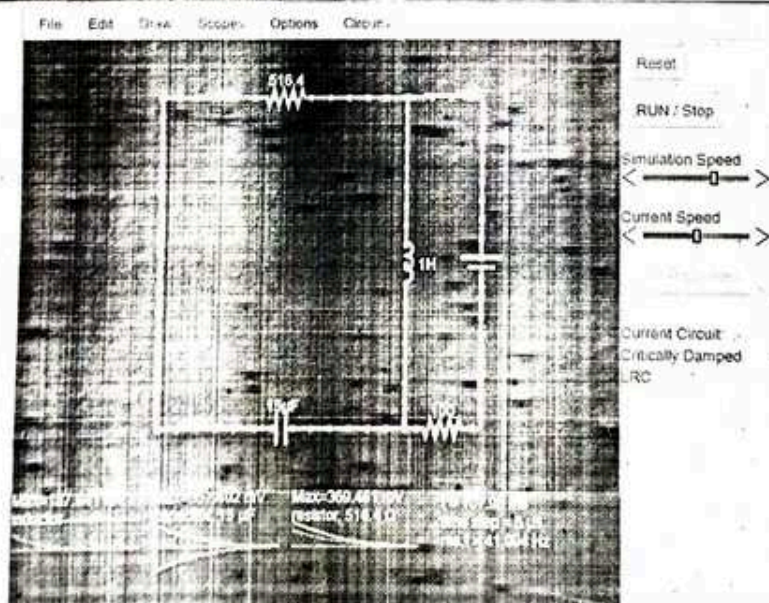
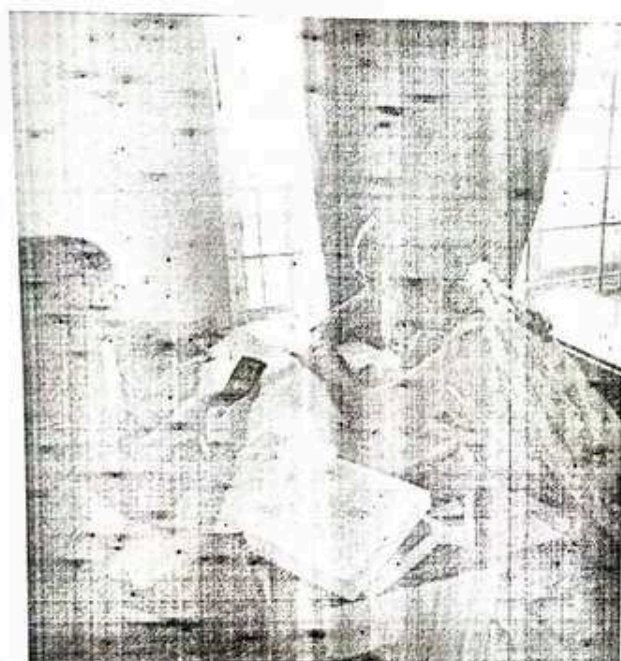
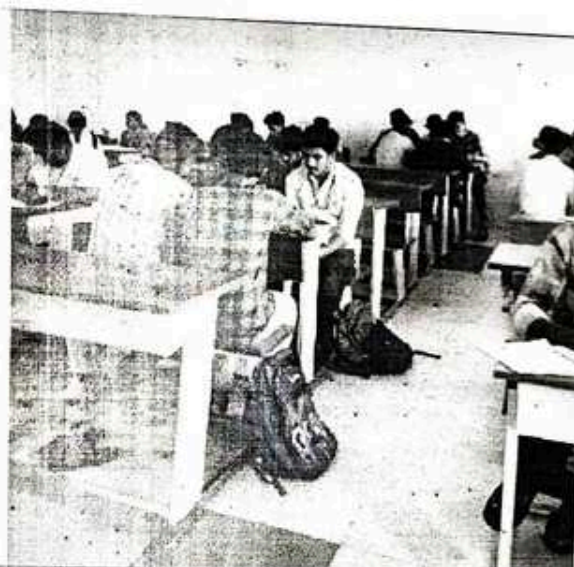
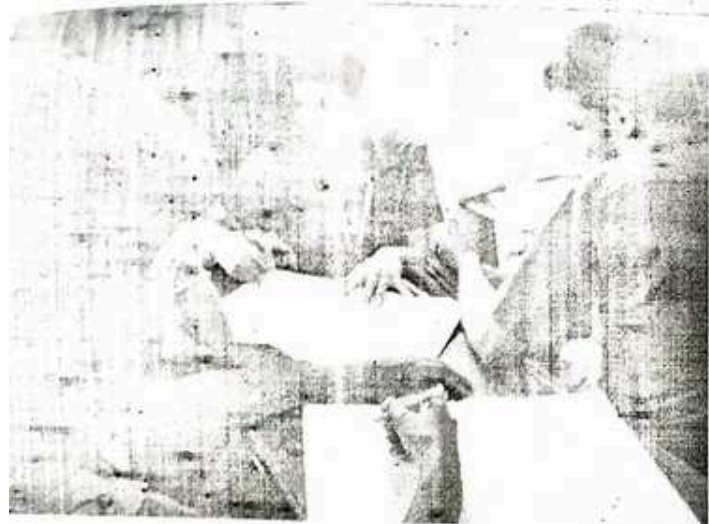
CO – PO / PSO mapping:

CO	PO1	PO8	PO9	PO10	PSO3
CO3	2	2	2	3	3

• **PO / PSO mapped:**

Innovative practice	PO1	PO8	PO9	PO10	PSO3
	3	2	2	3	3
Justification for correlation	Unit Step driven function requires a basic engineering fundamentals integration and differentiation. So this course outcome is highly correlated	TAI activity involves group discussion which apply ethical principles to derive the equation individually. The course outcome is moderately correlated	As the activity involves team work, the leadership qualities such as implementing norms of practice as a team to achieve a goal, listening to team members and treating them equally the Course outcome is mapped with moderate correlation	Through these activity students can communicate the technical and non-technical information with their team members effectively. Hence the course outcome is mapped with high correlation.	It is mapped with correlation level 3. Because to develop or analyze the response of RLC Circuits (Unit Step Function). Hence this activity is mapped with high correlation.

• Images / Screenshot of the practice:



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

Every student is actively involved in the learning process. Since each member of the group plays a crucial role in contributing to the overall understanding of the topic, there is a higher level of engagement. Students enjoy the challenge of working together to share the knowledge among them. Students appreciate the opportunity to learn from their peers.

- ❖ **Benefit of the practice:**

Every student got equal opportunity to come forward to take part in this activity. Students easily understand the response of Unit step driven function of RLC Circuits. This practice helped the students recall the derivation and solve the problem correctly. Students who formed groups were better able to understand concepts by exchanging ideas with one another.

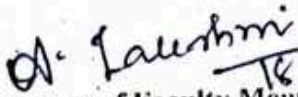
- ❖ **Challenges faced in implementation:**

- Faced issues while make the students to understand about TAI activity rules.
- This task has been scheduled for 45 minutes. Forming a group takes some time for the students.
- The main challenge faced is group conflict. Few Students are not ready to work together. Initially, the students not came up to involve in this activity voluntarily.

References:

TEXTBOOKS

1. Hayt Jack Kemmerly, Steven Durbin, "Engineering Circuit Analysis", Mc Graw Hill education, 9th Edition, 2018
2. Charles K. Alexander & Mathew N.O.Sadiku, "Fundamentals of Electric Circuits", Mc Graw- Hill, 2nd Edition, 2003
3. Joseph Edminister and Mahmood Nahvi, —Electric Circuits, Schaum's Outline Series, Tata McGraw Hill Publishing Company, New Delhi, Fifth Edition Reprint 2016.


Signature of Faculty Member


HOD



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering
Academic Year 2023 – 2024 (Even Semester)

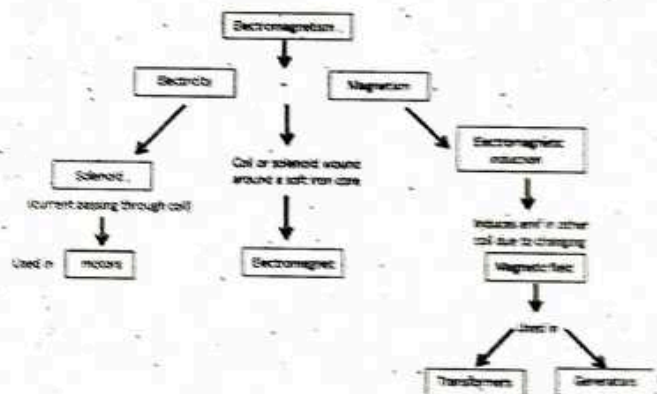
Degree, Semester & Branch: B.E. II & ECE A

Course Code & Title: EC3251 & Circuit Analysis

Name of the Faculty member (s): Dr.A.Lakshmi

Active Learning Description

- Unit / Topic: Unit-5/ Mind map
- Course Outcome: CO5
- Topic Learning Outcome: TLO 24.25
- Activity Chosen: Magnetically Coupled Circuits, mutual Inductance



A. Lakshmi
22/5/24
Signature of Faculty Member

[Signature]
HOD



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering

Academic Year 2023 – 2024 (Even Semester)

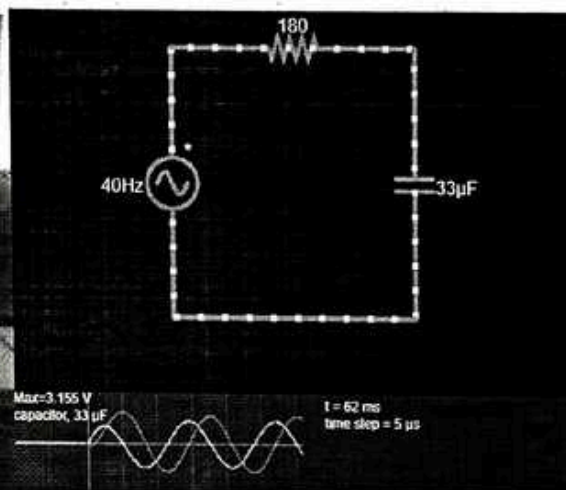
Degree, Semester & Branch: B.E, II & ECE A

Course Code & Title: EC3251 & Circuit Analysis

Name of the Faculty member (s): Dr.A.Lakshmi

Active Learning Description

- Unit / Topic: Unit-4 / The Unit- Step Function Driven RC Circuits
- Course Outcome: CO4
- Topic Learning Outcome: TLO 21
- Activity Chosen: Live demonstration



A. Lakshmi
16/5/24
Signature of Faculty Member

[Signature]
HOD



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering
Academic Year 2023 – 2024 (Even Semester)

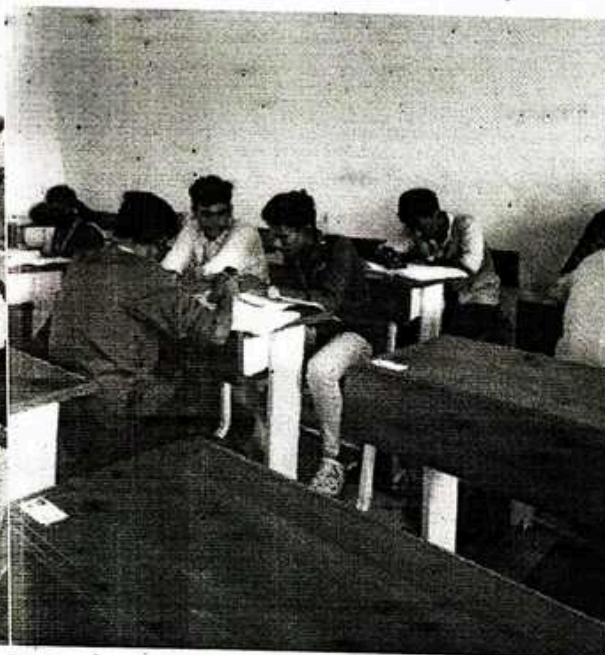
Degree, Semester & Branch: B.E, II& ECE A

Course Code & Title: EC3251 & Circuit Analysis

Name of the Faculty member (s): Dr.A.Lakshmi

Active Learning Description

- **Unit / Topic:** Unit-3 / Average Power, Apparent Power and Power Factor, Complex Power
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO 18
- **Activity Chosen:** Open book test



A. Lakshmi
7/5/24
Signature of Faculty Member

[Signature]
HOD



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering

Academic Year 2023 – 2024 (Even Semester)

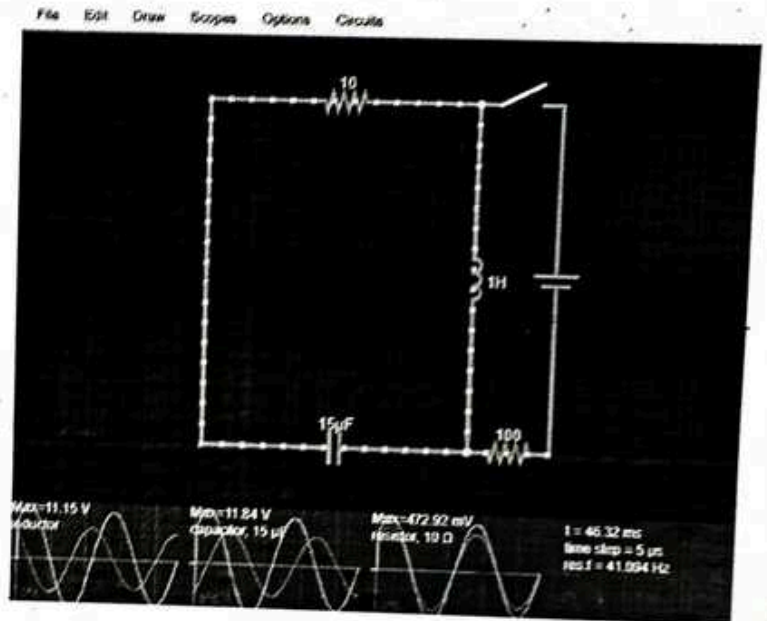
Degree, Semester & Branch: B.E., II & ECE, A

Course Code & Title: EC3251 & Circuit Analysis

Name of the Faculty member (s): Dr.A.Lakshmi

Active Learning Description

- Unit / Topic: Unit-3 / The Phasor, Phasor relationship for R, L
- Course Outcome: CO3
- Topic Learning Outcome: TLO 15
- Activity Chosen: Live Demonstration



A. Lakshmi
28/7/24

Signature of Faculty Member

HOD



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering

Academic Year 2023 – 2024 (Even Semester)

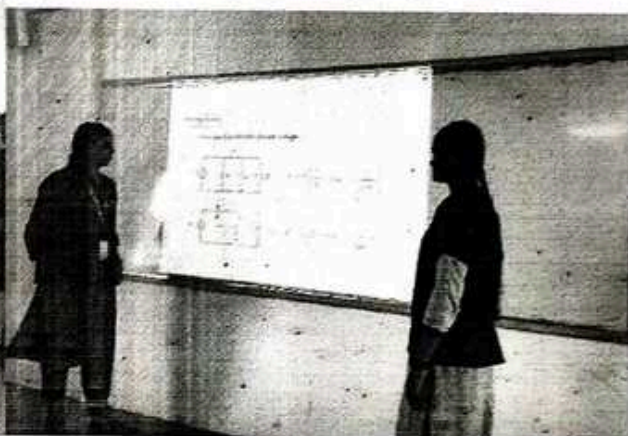
Degree, Semester & Branch: B.E, II & ECE A

Course Code & Title: EC3251 & Circuit Analysis

Name of the Faculty member (s): Dr.A.Lakshmi

Active Learning Description

- Unit / Topic: Unit-3 / The Complex Forcing Function
- Course Outcome: CO3
- Topic Learning Outcome: TLO 15
- Activity Chosen: Think-Aloud-Pair Problem Solving (TAPPS)



A. Lakshmi
17/4/24
Signature of Faculty Member


HOD



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: B.E, II & ECE A

Course Code & Title: EC3251 & Circuit Analysis

Name of the Faculty member (s): Dr.A.Lakshmi

Active Learning Description

- Unit / Topic: Unit-2 / Delta-Wye Conversion, Duals, Dual circuits
- Course Outcome: CO2
- Topic Learning Outcome: TLO 12,13
- Activity Chosen: Visible Quiz



Star to Delta Conversion Question 1:

The resistor value in a Y network that is equivalent to a Δ containing three resistors of $R \Omega$ each is:

1. $\frac{R}{3} \Omega$ each
2. $3 R^2 \Omega$ each
3. $\frac{R^2}{3} \Omega$ each
4. $3 R \Omega$ each

A. Lakshmi
Signature of Faculty Member

[Signature]
HOD



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering
Academic Year 2023 – 2024 (Even Semester)

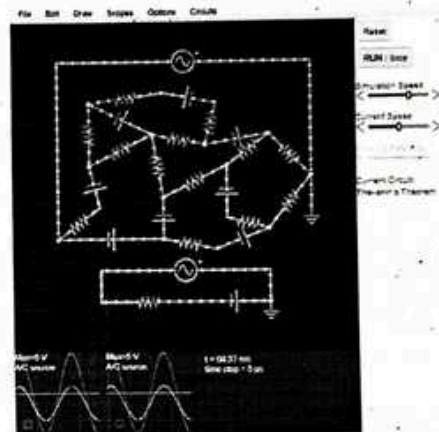
Degree, Semester & Branch: B.E, II & ECE A

Course Code & Title: EC3251 & Circuit Analysis

Name of the Faculty member (s): Dr.A.Lakshmi

Active Learning Description

- Unit / Topic: Unit-2 /Live demonstration
- Course Outcome:CO2
- Topic Learning Outcome: TLO 11
- Activity Chosen: Maximum Power Transfer theorem



A. Lakshmi
1/4/24
Signature of Faculty Member

[Signature]
HOD



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering

Academic Year 2023 – 2024 (Even Semester)

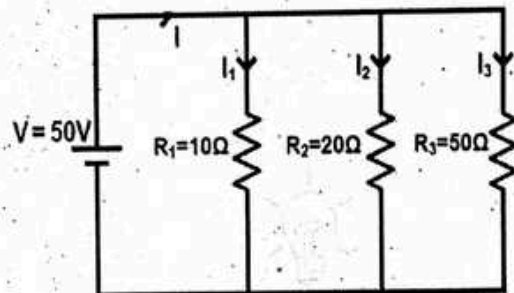
Degree, Semester & Branch: B.E, II & ECE A

Course Code & Title: EC3251 & Circuit Analysis

Name of the Faculty member (s): Dr.A.Lakshmi

Active Learning Description

- Unit / Topic: Unit-1 / Voltage and Current division
- Course Outcome: CO1
- Topic Learning Outcome: TLO 6
- Activity Chosen: Class Poll



A. Lakshmi
6/3/24
Signature of Faculty Member

[Signature]
HOD



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering
Academic Year 2023 – 2024 (Even Semester)

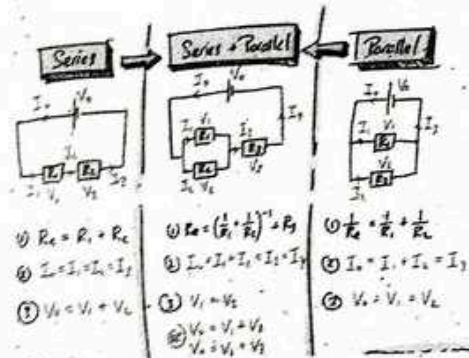
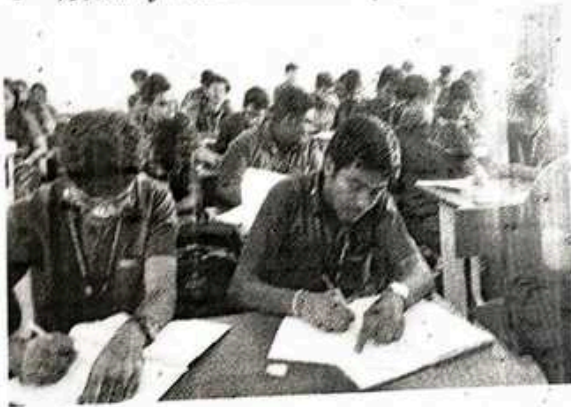
Degree, Semester & Branch: B.E, II & ECE A

Course Code & Title: EC3251 & Circuit Analysis

Name of the Faculty member (s): Dr.A.Lakshmi

Active Learning Description

- Unit / Topic: Unit-1 / The single Node – Pair Circuit, Series and Parallel Connected Independent Sources
- Course Outcome: CO1
- Topic Learning Outcome: TLO 4
- Activity Chosen: Concept Note



A. Lakshmi
Signature of Faculty Member 5/3/24

HOD



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering
Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch: B.E, II & ECE A

Course Code & Title: EC3251 & Circuit Analysis

Name of the Faculty member (s): Dr.A.Lakshmi

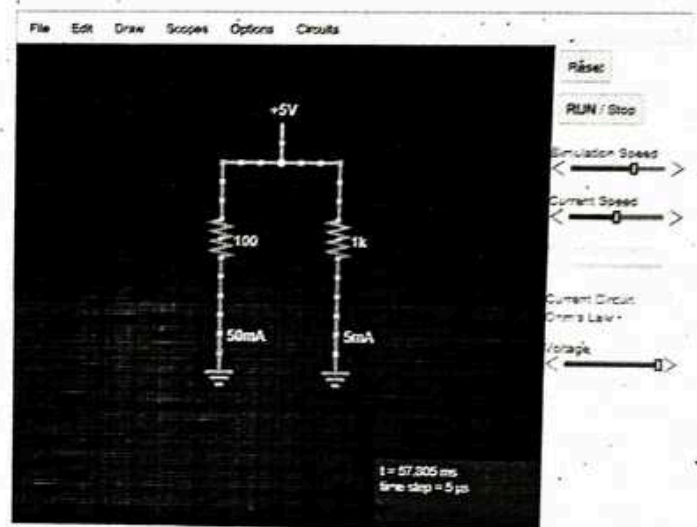
Active Learning Description

Unit / Topic: Unit-1 / Voltage and Current Sources, Ohms Law

Course Outcome: CO1

Topic Learning Outcome: TLO 2

Activity Chosen: Live demonstration



Dr. Lakshmi
4/3/24
Signature of Faculty Member

HOD